

SPOTIFY TRACKS PERFORMANCE ANALYSIS

Project by:

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DATA ANALYTICS

BATCH: 11.30 AM -1.30 PM

Project Overview:

This project analyzes Spotify tracks to understand song popularity, artist performance, and audio characteristics. The analysis explores relationships between popularity and musical features such as energy, danceability, duration, and tempo. The dashboard provides insights into top-performing artists, most popular tracks, song distribution, and factors that influence track popularity.

Business Problem:

Spotify hosts millions of songs, making it difficult for artists, music producers, and Spotify itself to understand which factors contribute to a song's popularity. Without data-driven insights, it is challenging to identify successful artists, track performance trends, and the audio characteristics that influence listener engagement.

Therefore, Spotify needs an analytical solution to examine song popularity, artist performance, and audio features such as energy,

danceability, duration, and tempo to support better content recommendations and music production decisions.

Dataset Summary:

No. of Rows: 1,30,987

No. of Columns: 17

Column Names:

artist_name	Name of the artist who performed the song
track_id	Unique identifier assigned to each song by Spotify.
track_name	Name of the song or track.
acousticness	Measures how acoustic (non-electronic) a song sounds, from 0 to 1.
danceability	Indicates how suitable a song is for dancing, from 0 to 1.
duration_ms	Length of the song measured in milliseconds.
energy	Represents the intensity and activity level of a song, from 0 to 1.
instrumentalness	Predicts whether a song contains no vocals, from 0 to 1.
key	The musical key in which the song is

	composed.
liveness	Detects the presence of a live audience in the recording, from 0 to 1.
loudness	Overall loudness of a song measured in decibels (dB).
mode	Indicates whether the song is in Major (1) or Minor (0) mode.
speechiness	Measures the amount of spoken words in a song, from 0 to 1.
tempo	Speed of the song measured in beats per minute (BPM).
time_signature	Number of beats in each musical measure of the song.
valence	Measures the musical positivity or happiness of a song, from 0 to 1.
popularity	Spotify popularity score of the song ranging from 0 to 100.

Data Understanding

- Identified column names and data types.
- Explored the dataset structure.
- Understood relationships between variables.
- Examined numerical and categorical fields.

Data Cleaning

Data Cleaning is done in pandas. Through pandas,

- Merged the datasets.

- Checked for missing values.
- Removed duplicate records.
- Verified data types.
- Checked unique values.
- Reviewed outliers and inconsistencies.
- Created Duration (Minutes) from duration_ms.

Exploratory Data Analysis

KPI Analysis

- Total Songs
- Total Artists
- Average Popularity
- Average Duration

Artist Analysis

- Top 10 Artists by Average Popularity
- Top 10 Artists by Number of Tracks

Track Analysis

- Top 10 Tracks by Popularity

Popularity Analysis

- Distribution of Song Popularity
- Relationship between popularity and energy
- Popularity by duration

Duration Analysis

- Duration in bins
- Popularity by duration
- Average duration in minutes

Outlier Analysis

- Duration Outliers
- Energy Outliers

Dashboard Development

An interactive Tableau dashboard was developed containing:

- KPI Cards
- Popular Artist
- Popular Track
- Popularity Distribution of songs
- Duration Popularity
- Top Artist and their Track
- Relationship between popularity and energy
- Interactive Filters

Key Findings

- The dataset contains 130,989 songs from 34,464 artists. This indicates a diverse and extensive music library for analysis.

- The average popularity score is 24.24 out of 100. This suggests that most songs in the dataset are not highly popular, while only a small percentage achieve high popularity.
- The average song duration is approximately 3.4 minutes. This indicates that most songs follow the standard commercial music length.
- Popularity Distribution is Uneven. The popularity distribution chart shows that a large number of songs have lower popularity scores. Highly popular songs represent only a small portion of the dataset.
- Some artists consistently achieve much higher popularity than others. The Top 10 Artists chart highlights the most successful artists in terms of average popularity.
- The Top 10 Tracks chart shows that only a small number of songs achieve very high popularity scores. Popularity is concentrated among a limited set of tracks.
- From the Popularity vs Energy scatter plot, highly popular songs can be found at different energy levels. Energy alone does not determine a song's popularity.
- The Popularity vs Duration chart suggests that both short and long songs can become popular. Duration is not a major factor driving popularity.
- The "Top Artists by Number of Tracks" chart shows that a few artists contribute a large volume of songs compared to others.

Conclusion

The analysis reveals that song popularity is highly concentrated among a small number of tracks and artists. While audio features such as energy and duration contribute to song characteristics, they do not independently determine popularity. The dashboard provides valuable insights into artist performance, track popularity, and overall music trends within the Spotify dataset.